

Swesan Pathmanathan

Canadian Citizen

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EDUCATION

McMaster University

Bachelor of Engineering, Software Engineering (Co-op)

Hamilton, ON

Sept 2021 – Apr 2026

TECHNICAL SKILLS

Languages: C, C++, Python, SQL

Systems: Linux, Bash, Windows, GitHub, Docker, ROCm, BMC, BIOS/UEFI

Software Engineering: APIs, client-server integration, debugging, test automation, production quality workflows, Cursor, GitHub Copilot

Databases: PostgreSQL, IBM DB2

WORK EXPERIENCE

Advanced Micro Devices (AMD) | Python, C/C++, ROCm, Linux, GitHub, Docker, post-silicon GPU, profiling workflows, APIs, firmware, BMC, BIOS/UEFI

Markham, ON

Datacenter GPU System Integration Engineer (Co-op)

May 2024 – Aug 2025

- Acted as rotating **test lead** for MI300X, MI325X, and MI350X post-silicon GPU validation, coordinating **100+ runs weekly** on **Linux** and **Windows** across HPC and ML workload gates.
- Coordinated feature enablement with firmware, validation, and infrastructure partners across MI300X, MI325X, and MI350X profiling and workload paths, keeping cross-team integration milestones aligned for release cycles.
- Debugged **BMC** and **BIOS/UEFI** issues across MI300X, MI325X, and MI350X racks; documented flash and bring-up steps reused on **20+ rack bring-ups**, **cutting repeat setup clarifications by about 40%**.
- Extended **Python** automation and an internal workload launcher with version-tracked scripts, shrinking diagnostic setup from **about 20 minutes to under 5 minutes** per run through reusable launch paths and AI-assisted edits in **Cursor** and **GitHub Copilot**.
- Triaged defects across vendor **GPU** stacks including ROCm, firmware, C/C++, and interconnects using logs; added **repeatable test scripts** so every handoff shipped the same repro bundle, which **cut reopened and duplicate defects on my queue by about 30%** by eliminating one-off retests.

PROJECTS

SufSort Sorting System | C, Bash, Linux

Nov 2025

- Wrote two sorting programs in **C** to compare how different algorithms behave on large data files of up to several hundred megabytes.
- Used **Bash** scripts on **Linux** to generate input data, run batch tests, and check sorted output, catching logic errors early in development.

Hitchly Ride Share Application | Node.js, PostgreSQL, Docker

Sept 2025 – Present

- Implemented CI and release checks with GitHub Actions and **Docker** for linting, tests, and pull request validation, strengthening production software quality gates for each backend change.
- Built and integrated backend **APIs** in **Node.js** with a **Docker**-containerized **PostgreSQL** service, supporting client-server workflows and feature integration across the application stack.

Ontario Hospital Association Database Design | SQL, IBM DB2

Sept 2025

- Authored SQL scripts implementing primary keys, foreign keys, and referential integrity constraints to ensure consistency across patient and record data.
- Developed a full database creation script compatible with **IBM DB2** CAS server environments for reproducible deployment.

RELEVANT COURSES

Performance Analysis Tools and Methods, Concurrent Systems and Synchronization, Computer Networks Security, Database Systems